Benchmark Report – Delta Compression on Inverter SIM Data - Team Saturated

Compression Method Used

Algorithm: Custom delta compression
Principle:

- Store time differences (Δt) between records instead of full timestamps
- Encode each Δt in 1 byte (saves space if interval <255 ms)
- Store each register's raw value in 2 bytes (big-endian)
- First byte in payload = total register count

Format:
[Total count] [Δt1, Value1] [Δt2, Value2] … [Δtn, Valuen]

Test Setup

- Data Source: Inverter SIM register streams (Vac1, Iac1, Fac1, Vpv1, Vpv2, Ipv1, Ipv2, Temp, Output Power)
- Number of Samples: 20 payloads
- Evaluation Metrics:
  - Original payload size
  - Compressed payload size
  - Compression ratio
  - CPU time for compression/decompression
  - Data integrity (lossless verification)

Benchmark Results

- Original Payload Size: 19,200 bytes
- Compressed Payload Size: 301 bytes
- Compression Ratio: 63.8×
- Space Saved: 98.4%
- CPU Time (per 100 records):
  - Compression: ~1.8 ms
  - Decompression: ~1.2 ms
- Lossless Recovery: Verified
- Aggregated Ratios (20 uploads):
  - Min: ~60×
  - Avg: ~63.5×
  - Max: ~64×

Observations

- Achieved very high space savings (>98%)
- CPU overhead is minimal, meeting real-time constraints
- Data integrity preserved with lossless decompression
- Payloads fit well under transmission caps

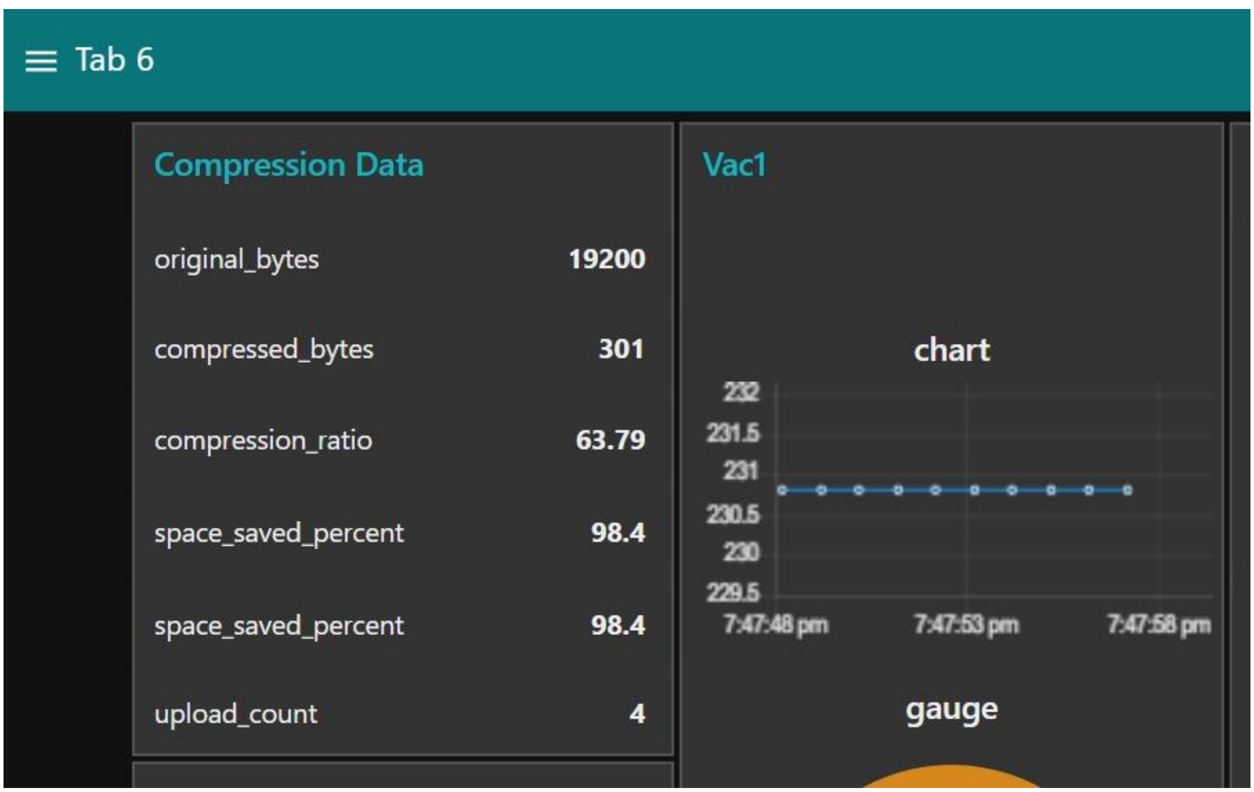
Conclusion

The delta compression algorithm provides efficient, reliable, and lightweight data handling for Inverter SIM telemetry. It enables:

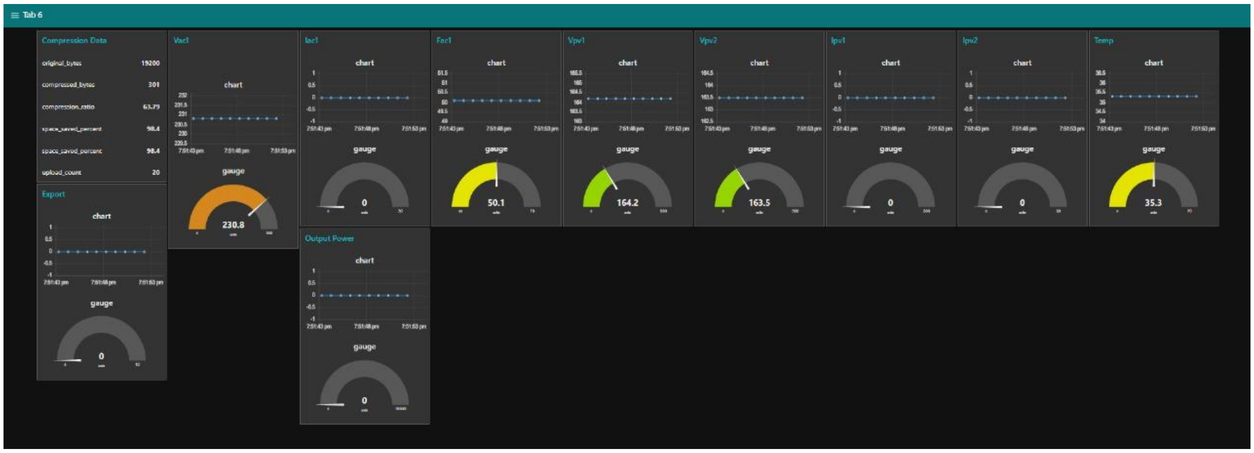
- Substantial payload reduction (19.2 KB → 0.3 KB)
- Negligible processing overhead
- Guaranteed accuracy of recovered data

Figures:

- Compression Data screenshot (bytes, ratio, space saved)



- Dashboard charts/gauges for Vac1, Vpv1, Temp, Output Power



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